



ASSOCIATION OF
EQUIPMENT MANUFACTURERS



EquipmentDealers
ASSOCIATION

**U.S. Copyright Office (Docket No. 2015-8)
Section 1201 Study:
Request for Additional Comment
September 27, 2016**

Joint Industry Comments
Respectfully Submitted October 26, 2016

The Association of Equipment Manufacturers (AEM) is a trade association that provides services on a global basis for companies that manufacture equipment, products and services used worldwide in the agriculture, construction, forestry, mining, and utility industries. AEM's membership includes more than 850 companies, many of which depend on the anti-circumvention provisions of the Digital Millennium Copyright Act (DMCA).¹ The DMCA protects the highly-creative, copyrighted digital software of AEM's members against unauthorized access, copying, and distribution.

The Equipment Dealers Association respectfully joins in the submission of these comments. Founded in 1900, the Equipment Dealers Association ("EDA") is non-profit trade organization representing approximately 4,000 retail dealers extensively engaged in the sale and service of agricultural, construction, industrial, forestry, outdoor power, lawn and garden, and/or turf equipment. EDA supports legislation or regulations which protect consumers and enhance consumer safety. EDA opposes legislation or regulations which allow the widespread release of proprietary equipment repair information to unqualified persons or entities which are not subject to uniform standards including training, safety, engineering and environmental. Such legislation or regulations could facilitate conduct which would put equipment operators and third parties at risk and put dealerships who invest in training, safety and liability coverage at an unfair economic disadvantage.

¹ 17 U.S.C. § 1201.

AEM and EDA submit these comments in connection with the Copyright Office's Section 1201 study to explain the importance of the technological measures in Section 1201(a) to the AEM's members and the public. The DMCA supports copyright by encouraging the development and availability of highly-valuable creative digital works of authorship in the products of AEM's members which are, in turn, distributed and serviced by EDA's members. The below section numbers correspond to the numbers of the specific questions on page 66296-66299 of the Section 1201 Study notice ("Study") published in the Federal Register.²

The Copyright Office seeks comments on several new proposed permanent exemptions, including computer programs that operate a machine or device. The following comments are limited to the subject matter specifically referenced.

1. *The Copyright Office suggests the possibility of "adoption of a permanent exemption for purposes of diagnosis, maintenance, and repair" and seeks "helpful starting points for further consideration of legislation." The proposed permanent exemption "could provide that a person who has lawfully obtained the right to use a computer program may circumvent a TPM controlling access to that program, so long as the circumvention is undertaken for purposes of diagnosis, maintenance, or repair." Study at p. 6.*

AEM and EDA oppose the enactment of any permanent exemption to circumvention for diagnosis, repair, maintenance or modification that would pose unnecessary risks to public safety, the environment and the economy. AEM and EDA caution against enactment of any permanent exemption where the copyrighted work, such as vehicle software or software updates, are commercially available products or services. If the works are already commercially available, the proposed permanent exemption would diminish or destroy the value of the vehicle software or software updates and unfairly disrupt the competitive landscape for both AEM and EDA members. Any permanent exemption must be limited to copyrighted works that are not commercially available or that are obsolete to reduce risks to public safety, the environment, and the economy.

A. Public Safety Risks: The proposed permanent exemption for vehicle diagnosis, repair, or maintenance of vehicle software poses significant public safety risks.

² Federal Register, Vol. 81, September 27, 2016, 66296-66299.

As Ambassador Michael Froman, U.S. Trade Representative, has emphasized, "counterfeit products, including semiconductors, automobile parts, and medicines, pose significant risks to consumer health and safety."³ Likewise, unauthorized modifications and/or repairs of vehicle software by untrained individuals raise similar safety concerns.

The proposed permanent exemption for vehicle diagnosis, repair or maintenance is a risky proposition as some vehicle owners may be willing to accept the risk of non-compliance with applicable safety and environmental requirements. These individuals include not only traditional "bad actors" but also individuals who simply lack an understanding of the risks that their non-compliant modification and/or repair may create for themselves or others.

Compounding these risks is the obvious inefficiency that this situation would create for regulators, such as the Environmental Protection Agency. It would be difficult and extremely costly to pursue enforcement of a large number of private vehicle owners who are readily able to hack, repair or modify vehicle software in non-compliant or otherwise dangerous ways. Conversely, the restriction of this information to a finite number of highly qualified and readily identifiable parties aids in the efficient and effective enforcement of safety and environmental regulations.

B. Environmental Risks: The proposed permanent exemption for diagnosis, repair, or maintenance of vehicle software poses significant environmental and air quality issues that can impact public health.

Providing increased access of the public to vehicle emissions control software has the potential to promote violations of EPA anti-tampering provisions and could result in vehicle owners altering software to improve vehicle performance at the expense of vehicle emissions, which is sometimes referred to as "engine chipping." Some engine chipping can relate to changing software programs or settings that are compliant with EPA certification to non-compliant software instructions that tend to violate emission regulations. AEM and EDA have serious concerns about both the potential harm to the environment and the potential liability exposure that their members may face with the widespread dissemination of this information to unqualified parties who may be willing accept the risk of non-compliance and, as discussed *supra*, may be difficult to hold accountable for violations.

³ See, e.g., Executive Office of the President of the United States, Ambassador Michael Froman, United States Trade Representative, 2015 Special 301 Report at p. 6.

C. Economic Risks: The proposed permanent exemption for vehicle diagnosis, maintenance, and repair (or modification) should not apply to commercially available software updates or to vehicle software where software updates are already available. The proposed permanent exemption for diagnosis, repair, or maintenance of vehicle software is contrary to sound economic policy. The general prohibition on circumvention of technological measures in section 1201 encourages innovation, prevents freeriding and supports a strong economy, particularly with respect to innovative vehicle software with expressive or creative content.

Many software updates in the equipment industry are commercially available to consumers through the authorized dealer network or directly from their equipment manufacturer. If the software is already commercially available, the fair use analysis of the software favors not providing a permanent exemption under section 1201. In particular, the fair use factor includes "the effect of the use upon the potential market for or value of the copyrighted work."⁴

Here, the proposed permanent exemption would impair the potential market for software updates or any diagnostic software that are commercially available from the equipment manufacturer and equipment dealers. Such software updates can provide improvements, maintenance or repairs, for example. Reputable manufacturers and equipment dealers have the economic burden of developing or providing tested software updates that comply with safety standards, such as the (Automotive Safety Integrity Level) ASIL standard, and all applicable state and federal emissions regulations. In contrast, the vehicle owner does not typically possess the engineering knowledge or resources to develop software code or updates that complies with safety requirements or environmental requirements.

What's more, if private equipment owners are permitted to publish or market alternate sources of software updates, which are untested with respect to safety or emissions requirements, such untested software would have an unfair price advantage (and a quality deficit) over tested alternatives and could easily displace software updates developed by reputable manufacturers and provided by authorized dealers to their customers. The untested vehicle software could degrade or destroy the market for software upgrade development by AEM's members and for sale by EDA's members. In the end, used vehicles for resale may include unknown software modifications or untested software updates that change machine

⁴ 17 U.S.C. § 107.

behavior in unpredictable ways creating unsafe situation(s) for consumers and new liability issues for equipment manufacturers and dealers.

Under current practices, equipment manufacturers provide software updates and patches to vehicle software through their dealers or directly to vehicle owners. There is no need for the vehicle owner to undertake the risk of endangering himself or herself, his or her family, passengers and bystanders by attempting to make repairs or modifications to vehicle software, or to procure such solutions from risky unknown sources that are not familiar with the design of a particular vehicle, or its software.

To the extent that the vehicle owners or third parties lack the resources or skill to develop and produce software that is compliant with voluntary safety standards (e.g., ASIL (Automotive Safety Integrity Level)), mandatory safety regulations and emissions regulations, public policy should favor manufacturers and dealers that comply with applicable laws, regulations and voluntary safety standards. Accordingly, it is recommended that the proposed expansion of circumvention measures does not apply where software updates are already commercially available.

Furthermore, the purpose of the DMCA's anti-circumvention provisions are in place to enable creative persons and businesses that operate in the U.S. to protect copyrights in digital works, such as music recordings, video recordings, and software, against unauthorized access and unauthorized copying and distribution.⁵ The DMCA helps mitigate against the threat of "massive piracy" of digital works due to "the ease with which [they] can be copied and distributed worldwide virtually instantaneously."⁶ Therefore, since 1998 when the DMCA was passed; software developers, distributors and manufacturers of equipment, which use embedded software, have relied upon anti-circumvention measures as a viable tool to prevent or discourage piracy and anti-competitive free riding on the creative authorship of copyright owners.

In conclusion, a permanent exemption of this nature would put equipment manufactures and authorized dealerships who invest in testing to ensure that software systems and updates provided to consumers are in compliance with both safety and environmental regulations at an unfair economic disadvantage.

⁵ 5 17 U.S.C. § 1201(b).

⁶ *Lexmark International, Inc. v. Static Control Components, Inc.*, 387 F.3d 522, 549 (6th Cir. 2004) (citing S. Rep. No. 105-190, at 8 (1998)).

